

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

UNIT II ELECTROSTATICS

Degree, Semester & Branch: IV Semester B.E. ECE A

Course Code & Title: EC8451 Electromagnetic Fields

Name of the Faculty member: Mrs.M.Sangeetha

Name of the Topic: Capacitance, Parallel capacitors, Cylindrical and spherical capacitors

Name of the Innovative Practice: Animated Video

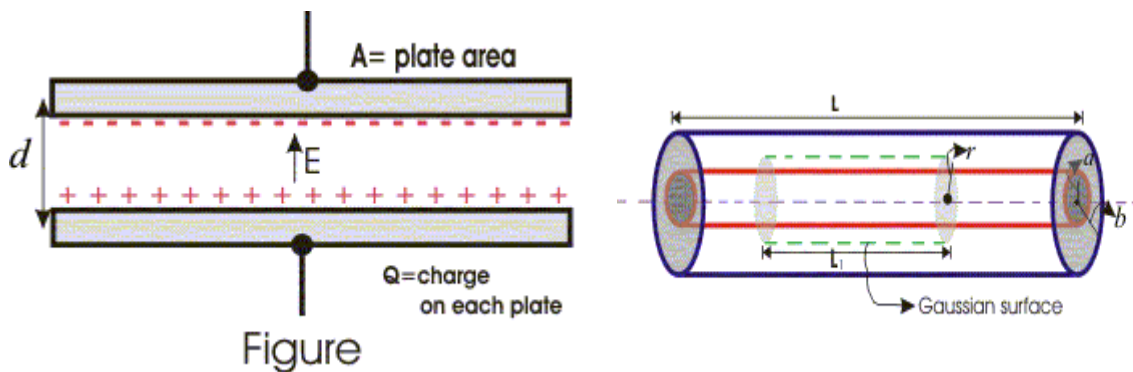
ICT Tool Used:- Smart Class Room

Description of Video Lecture:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

A parallel plate capacitor consists of two large plane parallel conducting plates separated by a small distance shown below in the figure.



A cylindrical capacitor is made up of a conducting cylinder or wire of radius a surrounded by another concentric cylindrical shell of radius b ($b > a$) shown in above figure.

A spherical capacitor consists of a solid or hollow spherical conductor of radius a , surrounded by another hollow concentric spherical of radius b shown below in figure



I have shown NPTEL Video Lecture series on “Electromagnetic Fields” by Dr. Harishankar Ramachandran, Department of Electrical Engineering, IIT Madras. In that Lecture No.14 describes about Capacitors. After shown this video, students can be easily differentiated the designing details of Parallel capacitors, Cylindrical and spherical capacitors.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	2	1					1	1		

CO2: The student will be able to describe the laws associated to static electric field and the properties of conductors and dielectrics.

References:

1. M.N.O. Sadiku and S.V. Kulkarni, Principles of electromagnetics, 6th ed., Oxford (Asian Edition), 2015
2. D.K. Cheng, Field and wave electromagnetics, 2nd ed., Pearson (India), 1989 (UNIT I, II, III IV,V)
3. <https://www.youtube.com/watch?v=bow4rworTk&list=PL1CE5B4FFFA997E5D&index=14>